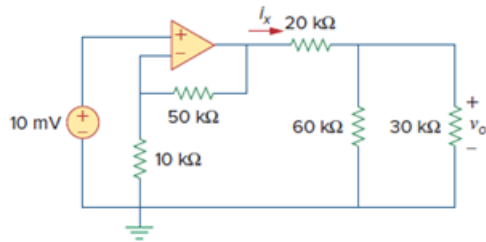


Problem

Calculate i_x and v_o in the circuit of Fig. 5.70. Find the power dissipated by the 60-k Ω resistor.

Figure 5.70:



Step-by-step solution

Step 1 of 5

Refer to Figure 5.70 in the textbook.

Draw the modified circuit diagram.

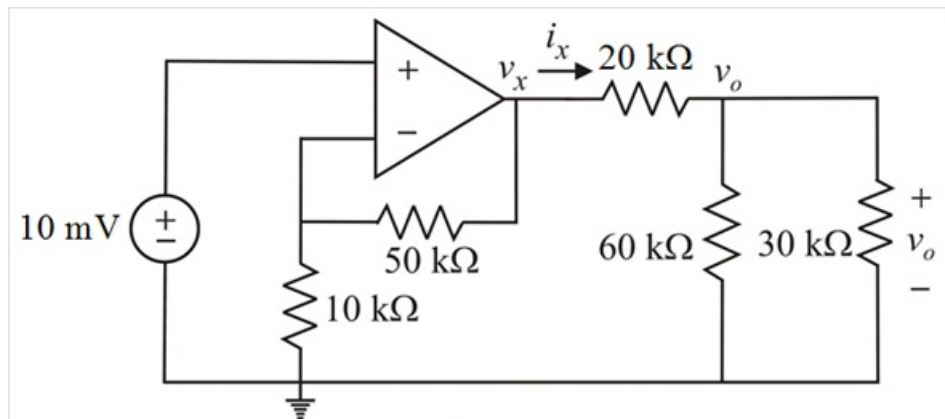


Figure 1

Step 2 of 5

The circuit is a non-inverting amplifier.

The output voltage of a non-inverting amplifier is,

$$v_x = \left(1 + \frac{R_f}{R_i} \right) v_i$$

Substitute $50 \text{ k}\Omega$ for R_f , $10 \text{ k}\Omega$ for R_i and 10 mV for v_i in the equation.

$$\begin{aligned} v_x &= \left(1 + \frac{50 \times 10^3}{10 \times 10^3} \right) (10 \times 10^{-3}) \\ &= (1 + 5) (10 \times 10^{-3}) \\ &= 60 \text{ mV} \end{aligned}$$

Step 3 of 5

Apply Kirchhoff's current law at node v_o .

$$\frac{v_o - v_x}{20 \times 10^3} + \frac{v_o}{60 \times 10^3} + \frac{v_o}{30 \times 10^3} = 0$$
$$v_o \left(\frac{1}{20 \times 10^3} + \frac{1}{60 \times 10^3} + \frac{1}{30 \times 10^3} \right) = \frac{v_x}{20 \times 10^3}$$
$$v_o \left(\frac{1}{20} + \frac{1}{60} + \frac{1}{30} \right) = \frac{v_x}{20}$$

Substitute 60 mV for v_x in the equation.

$$v_o \left(\frac{1}{20} + \frac{1}{60} + \frac{1}{30} \right) = \frac{60 \times 10^{-3}}{20}$$
$$v_o (0.1) = \frac{60 \times 10^{-3}}{20}$$
$$v_o = \frac{60 \times 10^{-3}}{20(0.1)}$$
$$v_o = 30 \text{ mV}$$

Thus, the value of output voltage, v_o is 30 mV.

Step 4 of 5

Determine the value of current, i_x .

$$i_x = \frac{v_x - v_o}{20 \times 10^3}$$

Substitute 30 mV for v_o and 60 mV for v_x in the equation.

$$i_x = \frac{(60 \times 10^{-3}) - (30 \times 10^{-3})}{20 \times 10^3}$$
$$= 1.5 \times 10^{-6}$$
$$= 1.5 \mu\text{A}$$

Thus, the value of current, i_x is 1.5 μA .

Step 5 of 5

Determine the power dissipated by the 60 k Ω resistor.

$$P_{60 \text{ k}\Omega} = \frac{v_o^2}{60 \times 10^3}$$

Substitute 30 mV for v_o in the equation.

$$P_{60 \text{ k}\Omega} = \frac{(30 \times 10^{-3})^2}{60 \times 10^3}$$
$$= \frac{900 \times 10^{-6}}{60 \times 10^3}$$
$$= 15 \times 10^{-9}$$
$$= 15 \text{ nW}$$

Thus, the power dissipated by the 60 k Ω resistor, $P_{60 \text{ k}\Omega}$ is 15 nW.